exercise-bank

Typst Package

Exercise Management & Banking System

A comprehensive solution for creating, organizing, and filtering exercises
Version 0.6.4
Nathan Scheinmann

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1. Introduction

`exercise-bank` is a Typst package for creating and managing exercises with solutions, metadata, filtering, and exercise banks. Perfect for teachers, textbook authors, and educational content creators.

1.1. Features

- Exercises with inline or deferred solutions
- Teacher corrections with flexible display modes
- Difficulty levels shown as badge colors, stars, or symbols
- Clickable links between exercises and their deferred corrections
- Split solution/correction placement, with an inline epigraph-style solution mode
- Chapter-prefixed numbering (“3.5”) and automatic end-of-chapter corrections
- Draft mode for work-in-progress documents
- Multiple display control options (what, where, which content)
- Metadata support (topic, level, author, competencies)
- Exercise banks: define once, display anywhere
- Powerful filtering by any criteria
- Automatic numbering with reset options
- Customizable labels (localization support)

1.2. Installation

Import the package in your Typst document:

```
#import "@preview/exercise-bank:0.6.4": exo, exo-setup
```

1.3. Quick Start

Code

```
#exo(
  exercise: [Solve the equation $2x + 5 = 13$.]
)
```

Preview

Exercise 1 Solve the equation $2x + 5 = 13$.

2. Basic Usage

2.1. Simple Exercise

Code

```
#exo(
  exercise: [Calculate $3 + 4 times 2$.]
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

2.2. Exercise with Solution

Code

```
#exo(
  exercise: [Calculate $3 + 4 times 2$.],
  solution: [$3 + 4 times 2 = 3 + 8 = 11$],
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

Solution 1 $3 + 4 \times 2 = 3 + 8 = 11$

2.3. Multiple Exercises

Exercises are automatically numbered:

Code

```
#exo(exercise: [Simplify $x^2 + 2x + 1$.])
#exo(exercise: [Factor $x^2 - 4$.])
#exo(exercise: [Solve $2x - 6 = 0$.])
```

Preview

Exercise 1 Simplify $x^2 + 2x + 1$.

Exercise 2 Factor $x^2 - 4$.

Exercise 3 Solve $2x - 6 = 0$.

3. Display Control

The package uses three parameters to control display behavior.

3.1. `display` - What to Display

Controls what content is displayed:

- `"both"` (default) - Show exercises and solutions/corrections
- `"ex"` - Show only exercises
- `"sol"` - Show only solutions/corrections

Code

```
#exo-setup(display: "ex")  // Exercises only

#exo(
  exercise: [Solve $x + 3 = 7$.],
  solution: [$x = 4$],  // Hidden
)
```

Preview

Exercise 1 Solve $x + 3 = 7$.

3.2. `corr-display` - Which Content to Show

Controls whether to show solutions or corrections:

- `"solution"` (default) - Show solution content
- `"correction"` - Show correction content
- `"mixed"` - Default to solution, but use correction for exercises with `show-corr: true`

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Simplify $2x + 3x$.],
  correction: [
    $2x + 3x = 5x$
    (Combine like terms)
  ],
)
```

Preview

Exercise 1 Simplify $2x + 3x$.

Correction 1 $2x + 3x = 5x$ (Combine like terms)

3.3. `corr-loc` - Where to Display

Controls where solutions/corrections appear:

- `"after"` (default) - Immediately after each exercise
- `"pagebreak"` - With a page break between
- `"end-section"` - Collected at section end
- `"end-chapter"` - Collected at chapter end

Code

```
#exo-setup(corr-loc: "end-section")

#exo(exercise: [Exercise 1], solution: [Answer 1])
#exo(exercise: [Exercise 2], solution: [Answer 2])

#exo-print-solutions()  // Print collected solutions
```

Preview

Exercise 1 Exercise 1

Exercise 2 Exercise 2

Solutions

Solution 1 Answer 1

Solution 2 Answer 2

4. Corrections (Teacher Version)

Corrections are detailed solutions for teachers, including pedagogical notes and teaching tips.

4.1. Exercise with Correction

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Solve $x^2 = 9$.],
  correction: [
    *Teacher notes:*
    $x = plus.minus 3$

    Common mistake: forgetting the negative root.
  ],
)
```

Preview

Exercise 1 Solve $x^2 = 9$.

4.2. Solutions Only (Answer Key)

Show only solutions for student answer keys:

Code

```
#exo-setup(display: "sol")

#exo(
  exercise: [This is hidden],
  solution: [Only this solution is shown],
)
```

Preview

Solution 1 Only this solution is shown

4.3. The `sol-in-corr` Flag

When `corr-display: "correction"`, both correction AND solution are shown by default. Use `sol-in-corr: true` on an exercise to indicate the solution is already embedded in the correction (avoiding duplication):

Code

```
#exo-setup(corr-display: "correction")

// Without sol-in-corr: both correction and solution shown
#exo(
  exercise: [Problem A],
  correction: [Teacher notes only],
  solution: [$x = 5$],
)

// With sol-in-corr: only correction shown
#exo(
  exercise: [Problem B],
  correction: [
    *Solution:* $x = 3$

    _Teaching tip: Watch for sign errors._
  ],
  solution: [$x = 3$],
  sol-in-corr: true,  // Solution already in correction
)
```

Preview

Exercise 1 Problem A

Exercise 2 Problem B

4.4. Mixed Display Mode

Use `corr-display: "mixed"` to default to solutions while showing corrections for specific exercises:

Code

```
#exo-setup(corr-display: "mixed")

#exo(
  exercise: [Simple problem],
  solution: [Quick answer],
  correction: [Detailed explanation],
)

#exo(
  exercise: [Complex problem],
  solution: [Answer],
  correction: [Step-by-step solution],
  show-corr: true,  // Shows correction
)
```

Preview

Exercise 1 Simple problem

Exercise 2 Complex problem

5. Draft Mode

Show placeholders for incomplete exercises during document preparation.

5.1. With Draft Mode ON

Code

```
#exo-setup(
  draft-mode: true,
  solution-placeholder: [_[To be written]_],
)

#exo(
  exercise: [Solve $x + 5 = 12$],
  solution: [],  // Empty - shows placeholder
)
```

Preview

$\boxed{\textbf{Exercise 1}}$ Solve $x + 5 = 12$

5.2. With Draft Mode OFF (Default)

Empty solutions show minimal space without placeholders:

Code

```
#exo-setup(draft-mode: false)

#exo(
  exercise: [Solve $x + 5 = 12$],
  solution: [],  // Empty - no placeholder
)
```

Preview

$\boxed{\textbf{Exercise 1}}$ Solve $x + 5 = 12$

6. Exercise Banks

Define exercises once, display them anywhere. Perfect for creating reusable exercise collections.

6.1. Defining Bank Exercises

Use `exo-define` to register exercises without displaying them:

```
#exo-define(
  id: "quad-1",
  exercise: [Solve $x^2 - 5x + 6 = 0$.],
  topic: "quadratics",
  level: "1M",
  solution: [$x = 2$ or $x = 3$],
)

#exo-define(
  id: "geom-1",
  exercise: [Find the area of a circle with radius 5.],
  topic: "geometry",
  level: "1M",
  solution: [$A = 25pi$],
)
```

6.2. Displaying Bank Exercises

Use `exo-show` to display a specific exercise:

Code

```
#exo-show("quad-1")
```

Preview

Exercise 1 Solve $x^2 - 5x + 6 = 0$.

6.3. Selecting Multiple Exercises

Use `exo-select` with filters:

```
// All quadratics exercises
#exo-select(topic: "quadratics")

// Level 1M exercises only
#exo-select(level: "1M")

// Multiple topics
#exo-select(topics: ("quadratics", "geometry"))

// Limit count
#exo-select(topic: "algebra", max: 5)

// Custom filter
#exo-select(where: ex => ex.metadata.level == "hard")
```

7. Metadata and Filtering

7.1. Adding Metadata

Tag exercises for organization and filtering:

```
#exo(
  exercise: [Solve $x + 1 = 5$.],
  topic: "algebra",
  level: "easy",
  authors: ("Prof. Smith",),
)
```

7.2. Filtering Exercises

Display only exercises matching criteria:

```
#exo-filter(topic: "algebra")
#exo-filter(level: "easy")
#exo-filter(topic: "algebra", level: "hard")
```

8. Competency Tags

Tag exercises with competencies and display them visually.

Code

```
#exo-setup(show-competencies: true)

#exo-define(
  id: "comp-ex",
  exercise: [Solve and explain your reasoning.],
  competencies: ("C1.1", "C2.3", "C4.1"),
  solution: [Detailed solution here],
)

#exo-show("comp-ex")
```

Preview

Exercise 1 Solve and explain your reasoning.

C1.1 C2.3 C4.1

8.1. Filter by Competency

```
#exo-select(competency: "C1.1")
#exo-select(competencies: ("C1.1", "C2.3"))
```

9. Configuration

9.1. Global Setup

Use `exo-setup` to configure defaults:

```
#exo-setup(
  // Display control
  display: "both",               // "ex", "sol", "both"
  corr-display: "solution",    // "solution", "correction", "mixed"
  corr-loc: "after",           // "after", "pagebreak", "end-section", "end-chapter"
  // Labels
  exercise-label: "Exercise",
  solution-label: "Solution",
  correction-label: "Correction",
  // Counter behavior
  counter-reset: "section",   // "section", "chapter", "global"
  // Display options
  show-id: false,
  show-competencies: false,
  // Draft mode
  draft-mode: false,
  // Badge styling
  badge-style: "box",
  badge-color: black,
)
```

9.2. Badge Styles

Change the visual style of exercise badges:

```
// Circled number (no label text)
#exo-setup(badge-style: "circled")

// Filled circle with white number
#exo-setup(badge-style: "filled-circle", badge-color: rgb("#2563eb"))

// Pill-shaped badge
#exo-setup(badge-style: "pill")

// Arrow tag
#exo-setup(badge-style: "tag", badge-color: rgb("#1e40af"))
```

Available styles: `"box"` (default), `"circled"`, `"filled-circle"`, `"pill"`, `"tag"`, `"margin"`, `"border-accent"`, `"underline"`, `"rounded-box"`, `"header-card"`

9.3. Localization

Change labels for different languages:

Code

```
// French
#exo-setup(
  exercise-label: "Exercice",
  solution-label: "Solution",
  correction-label: "Corrigé",
)
```

Preview

French: Exercice, Solution, Corrigé
German: Aufgabe, Lösung
Spanish: Ejercicio, Solución

9.4. Counter Reset Options

Control when numbering resets:

```
#exo-setup(counter-reset: "section")  // Reset each section
#exo-setup(counter-reset: "chapter")  // Reset each chapter
#exo-setup(counter-reset: "global")   // Never reset
```

Use hooks to trigger resets:

```
= New Section
#exo-section-start()  // Resets if counter-reset: "section"

= New Chapter
#exo-chapter-start()  // Resets if counter-reset: "chapter"
```

9.5. Advanced Exercises

Mark exercises as advanced to display a visual cue before the label. The default symbol is *.

Code

```
#exo(
  exercise: [A challenging problem.],
  advanced: true,
)
```

Preview

∗ Exercise 1 A challenging problem.

Customize the symbol:

Code

```
#exo-setup(advanced-symbol: sym.dagger)
#exo(
  exercise: [Advanced with dagger.],
  advanced: true,
)
```

Preview

† Exercise 1 Advanced with dagger.

Disable with `advanced-symbol: none`.

9.6. Optional Exercises

Mark exercises as optional to display the optional-star icon before the label. Students know they can skip these.

Code

```
#exo(
  exercise: [A bonus problem.],
  optional: true,
)
```

Preview

⋆ Exercise 1 A bonus problem.

Customize or disable the symbol via `exo-setup`:

```
#exo-setup(optional-symbol: [⭐])   // custom
#exo-setup(optional-symbol: none)    // disable
```

9.7. Correction-Given Exercises

The dumbbell icon (`corr-given: true`) signals that the printed correction will be handed out to students, so they can work independently with the answer key.

Code

```
#exo(
  exercise: [Factor $x^2 - 9$.],
  solution: [$(x-3)(x+3)$],
  corr-given: true,
)
```

Preview

Exercise 1 Factor $x^2 - 9$.

Customize via `exo-setup(corr-given-symbol: ...)` or disable with `none`.

10. QR Codes

Attach a QR code to any exercise (e.g. linking to a video correction or an online resource). Pass a URL string — the code is generated with tiaoma — or ready-made content via the `qr` parameter.

Code

```
#exo(
  exercise: [Factorise $x^2 - 9$.],
  solution: [$(x-3)(x+3)$],
  qr: "https://example.com/eq1",
)
```

Preview

Exercise 1 Factorise $x^2 - 9$.

The QR code is placed automatically based on the active badge style:

- **Badge styles** (`box`, `circled`, `filled-circle`, `pill`, `tag`, `margin`): the QR sits below the badge in the label margin.
- **Full-width styles** (`border-accent`, `underline`, `rounded-box`, `header-card`): the exercise content wraps around the QR at the top right.

Solution and correction boxes can carry their own separate QR codes via `qr-sol` and `qr-corr`.

10.1. Global QR Settings

Control QR appearance globally via `exo-setup`:

Code

```
#exo-setup(
  qr-size: 1.5cm,
  qr-color: rgb("#1e3a8a"),
  qr-caption: [Correction],
  show-qr: true,
)
```

Preview

Use `show-qr: false` to suppress all QR codes (e.g. for a print version) without removing the URLs from bank definitions.

10.2. QR Placement Modes

`qr-position` controls how the exercise body flows around the QR code for full-width badge styles:

Value	Behavior
`"auto"` (default)	Placed per badge style: label margin for badge styles, wrapped for full-width styles

`"wrap"`	Always wrap the exercise text around the QR at the top right, using wrap-it
`"tasks"`	Overlay the QR at the top right without reserving flow height; a taskize `#tasks` body inside the exercise narrows its own top rows to flow around the QR instead of the whole block being pushed below it

`"tasks"` is opt-in and only takes effect when the exercise body actually contains a `#tasks(...)` call from taskize 0.2.8 or newer — the two packages coordinate purely through a shared state key (`taskize-wrap-zone`), with no import dependency in either direction. Bodies without a `#tasks` block ignore the overlay zone and render as if `qr-position` were unset.

Code

```
#exo-setup(qr-position: "tasks")
#exo(
  exercise: tasks(
    [First short task.],
    [Second short task.],
    [Third short task.],
  ),
  qr: "https://example.com/quiz",
)
```

Preview

11. Difficulty Levels

Tag each exercise with a `difficulty:` level (built-in scale: 1 to 5). By default the exercise badge takes the level color:

Code

```
#exo(exercise: [Introductory.],
  difficulty: 1)
#exo(exercise: [Exam-type.],
  difficulty: 3)
#exo(exercise: [Advanced.],
  difficulty: 4)
```

Preview

Exercise 1 Introductory.

Exercise 2 Exam-type.

Exercise 3 Advanced.

11.1. Stars and Symbols

`difficulty-display: "stars"` shows 1–5 small stars; `"symbols"` shows one drawn icon per level (seedling, pencil, target, mountain, star). Both are placed below the badge by default so the badge stays compact (`difficulty-position: "badge"` puts them inline).

Code

```
#exo-setup(
  difficulty-display: "stars")
#exo(exercise: [Three stars.],
  difficulty: 3)

#exo-setup(
  difficulty-display: "symbols")
#exo(exercise: [Mountain.],
  difficulty: 4)
```

Preview

Exercise 1 Three stars.

★★★

Exercise 2 Mountain.

11.2. Custom Scale and Filtering

The scale is a dictionary mapping any key to a color and/or symbol:

```
#exo-setup(difficulty-scale: (
  "easy": (color: rgb("#00897b")),
  "hard": (color: rgb("#e65100"), symbol: [🔥]),
))
#exo(exercise: [...], difficulty: "hard")

// Filtering
#exo-select(difficulty: 3)
#exo-select(difficulties: (1, 2))
#exo-count(difficulty: 4)
```

Difficulty combines well with the `optional` marker: encode every exercise's level, and mark the mandatory ones with `optional: false`.

12. Solution Placement and Links

12.1. Automatic End-of-Chapter Corrections

With `corr-loc: "end-chapter"` the corrections are only *collected*; print them by calling `#exo-chapter-end()` where they should appear, or wrap the document with `exo-auto-chapter` to do it automatically before each new level-1 heading and at the end of the document:

```
#exo-setup(corr-loc: "end-chapter", counter-reset: "chapter")
#show: exo-auto-chapter

= Chapter 1
#exo(exercise: [...], solution: [...])

= Chapter 2  // <- Chapter 1 solutions print just before this title
```

12.2. Separate Solution and Correction Placement

`sol-loc` controls where *solutions* go, independently of `corr-loc` (default `auto` = follow it). Typical setup: short answer under the statement, full correction at the end of the chapter:

```
#exo-setup(
  corr-display: "correction",  // show correction and solution
  sol-loc: "after",            // answer below the statement
  corr-loc: "end-chapter",     // correction at chapter end
  solution-style: "inline",    // epigraph-like, no badge
)
```

12.3. Inline Solution Style

`solution-style: "inline"` renders the solution as a short rule + content right under the statement, with a small margin label (customizable via `inline-label`, `none` to hide; rule length via `inline-rule-length`):

Code

```
#exo-setup(
  solution-style: "inline")
#exo(
  exercise: [Solve
    $x^2 - 5x + 6 = 0$.],
  solution: [$x in {2, 3}$],
)
```

Preview

$\boxed{\textbf{Exercise 1}}$ Solve $x^2 - 5x + 6 = 0$.

Solution

$x \in \{2, 3\}$

12.4. Clickable Exercise ↔ Correction Links

When corrections are deferred, `link-solutions: true` adds a clickable arrow icon next to the exercise badge that jumps to the correction, and a back-link on the correction. With `link-style: "page"` the exercise instead shows a textbook-style clickable reference ("Solution p. 30") at the top right of the statement:

```
#exo-setup(corr-loc: "end-chapter", link-solutions: true)
// or, textbook style:
#exo-setup(corr-loc: "end-chapter", link-solutions: true,
  link-style: "page")
```

Customize with `link-icon` / `backlink-icon` (icons) or `page-ref-color` / `page-ref-format` (page reference).

12.5. Chapter-Prefixed Numbering

`number-prefix: "heading"` prefixes displayed numbers with the current level-1 heading number, e.g. exercise 5 of chapter 3 shows as "3.5" (separator via `number-separator`):

```
#set heading(numbering: "1.")
#exo-setup(number-prefix: "heading", counter-reset: "chapter")
#show: exo-auto-chapter
```

`number-prefix` also accepts a counter or a function `() => value`, for heading packages that keep their own chapter counter instead of `counter(heading)`. With beautitled 0.3.0+ the native counter is kept in sync, so `number-prefix: "heading"` works directly; for earlier versions (or with `enable-parts`, where the first heading level is the part) use beautitled's exported counter:

```
#import "@preview/beautitled:0.2.7": beautitled-init, chapter-counter
#show: beautitled-init
#exo-setup(number-prefix: chapter-counter, counter-reset: "chapter")
#show: exo-auto-chapter
```

With beautitled's direct `#chapter(...)` calls (no native headings), wrap the call instead of using `exo-auto-chapter`:

```
#let chapitre(..args) = {
  exo-chapter-end(); chapter(..args); exo-chapter-start()
}
```

13. Visual Styles

Set the badge style with `exo-setup(badge-style: "...")`.

13.1. Box (Default)

Code

```
#exo-setup(badge-style: "box")
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

$\boxed{\textbf{Exercise 1}}$ Solve $x + 3 = 7$

13.2. Circled

Code

```
#exo-setup(badge-style: "circled")
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

1 Solve $x + 3 = 7$

13.3. Filled Circle

Code

```
#exo-setup(
  badge-style: "filled-circle",
  badge-color: rgb("#2563eb"),
)
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

1 Solve $x + 3 = 7$

13.4. Rect and Filled Rect

Compact number-only rectangles – a minimal alternative when circles look too large in your font:

Code

```
#exo-setup(badge-style: "rect")
#exo(exercise: [Solve $x + 3 = 7$])
#exo-setup(
  badge-style: "filled-rect",
  badge-color: rgb("#1a4d8f"),
)
#exo(exercise: [Solve $2x = 10$])
```

Preview

1 Solve $x + 3 = 7$

2 Solve $2x = 10$

13.5. Custom Badge Function

For full control, pass a function `(label, number, font-size, color, is-solution) => content` as `badge-style`:

Code

```
#exo-setup(badge-style:
  (label, number, font-size,
   color, is-solution) => {
    box(stroke:
      (bottom: 1.5pt + color),
      inset: (x: 4pt, y: 3pt),
      text(weight: "bold",
        size: font-size,
        fill: color)[#number.])
  })
#exo(exercise: [Solve $x^2 = 4$])
```

Preview

1. Solve $x^2 = 4$

13.6. Pill

Code

```
#exo-setup(badge-style: "pill")
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1 Solve $x + 3 = 7$

13.7. Tag

Code

```
#exo-setup(
  badge-style: "tag",
  badge-color: rgb("#1e40af"),
)
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1 Solve $x + 3 = 7$

13.8. Border Accent

Code

```
#exo-setup(
  badge-style: "border-accent",
  badge-color: rgb("#3b82f6"),
)
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.9. Underline

Code

```
#exo-setup(badge-style: "underline")
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.10. Rounded Box

Code

```
#exo-setup(badge-style: "rounded-box")
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.11. Header Card

Code

```
#exo-setup(badge-style: "header-card", badge-color: rgb("#3b82f6"))
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.12. Margin

The label appears flush-right in a fixed-width left column; content flows in the right column. Solutions are shown in a compact bordered block.

Code

```
#exo-setup(badge-style: "margin")
#exo(
  exercise: [Solve $x + 3 = 7$],
  solution: [$x = 4$],
)
```

Preview

Exercise 1: Solve $x + 3 = 7$

Solution 1

$x = 4$

13.13. Badge Position

Every style above places the badge in its own column on the left, so the statement is indented for the whole height of the exercise. In a narrow measure – two-column layouts especially – that column costs its width on **every** line, and it gets worse when the statement holds an enumeration, whose own indent stacks on top of it.

`badge-position: "above"` puts the badge alone on a header line and lets the statement run the full width underneath. It is independent of `badge-style`, so every badge look keeps working:

Code

```
#exo-setup(badge-style: "filled-circle")
#exo(exercise: [...])            // badge-position: "margin" (default)

#exo-setup(badge-position: "above")
#exo(exercise: [...])            // same badge, full-width statement
```

Preview

badge-position: "margin"

1 A floor is covered with 500 square tiles. Tiles 5 cm longer and wider would have taken 320 of them.

1. Express the side length of the first tiles.
2. How many tiles of side $x + 5$ would be needed?

badge-position: "above"

2

A floor is covered with 500 square tiles. Tiles 5 cm longer and wider would have taken 320 of them.

1. Express the side length of the first tiles.
2. How many tiles of side $x + 5$ would be needed?

With `link-style: "page"` the "Solution p. 34" reference moves onto the badge line as well, instead of being wrapped into the top right of the statement.

13.14. Badge Size

Every badge shape ships with its own paddings and corner radius, tuned for a 12pt label. `badge-scale` multiplies them: below 1 the badge closes in on the number, above 1 it grows into a block. The label text keeps the size given by `label-font-size` – scaling the badge changes the box, not the digits, so lower both together for a genuinely discreet badge.

Code

```
#exo-setup(badge-style: "filled-rect")

#exo-setup(badge-scale: 0.5)   // tight
#exo(exercise: [...])

#exo-setup(badge-scale: 1.0)   // default
#exo(exercise: [...])

#exo-setup(badge-scale: 1.6)   // roomy
#exo(exercise: [...])
```

Preview

1

Badge at scale 0.5.

2

Badge at scale 1.

3

Badge at scale 1.6.

For full control, `badge-pad-x`, `badge-pad-y` and `badge-radius` replace the shape's own values outright (and ignore `badge-scale`). The badge height is always `label-font-size + 2 * badge-pad-y`, so these two knobs also set how tall the badge is:

```
#exo-setup(
  badge-style: "pill",
  label-font-size: 9pt,
  badge-pad-x: 6pt,     // horizontal padding
  badge-pad-y: 2pt,     // -> 9pt + 4pt = 13pt tall
  badge-radius: 6.5pt,  // half the height: still a pill, just a smaller one
)
```

The margin reserved for the badge column follows the badge as it is resized, so `badge-position: "margin"` keeps its alignment without touching `margin-position`. Custom badge functions keep the documented five-argument contract and are left alone by all four settings – a function draws its own geometry.

14. Multi-Column Layouts

Exercise sheets often read better on two columns, statements and solutions flowing side by side. Typst's built-in `columns` does the flowing, but it draws no separator and it leaves the badge column at its full width, which eats a large share of a narrow measure. The package wraps both concerns.

14.1. A Whole Two-Column Document

`exo-page-columns` is a show rule: it switches the page to `count` columns, draws an optional vertical rule in the middle of every gutter, and (unless told otherwise) moves the badges above the statements so they cost no column width.

```
#import "@preview/exercise-bank:0.6.4": exo, exo-page-columns

#show: exo-page-columns.with(count: 2, rule: 0.5pt + gray)

#exo(exercise: [...], solution: [...])
#exo(exercise: [...], solution: [...])
```

Because these are page columns, the text flows from page to page as usual and the rule is redrawn on every page. Put the show rule after `#set page(...)`: it reads the page geometry to place the rule inside the text area, and it resolves `auto` margins the same way Typst does.

> **Order matters with `exo-auto-chapter`.** Show rules nest in the order they are written, so `exo-page-columns` must come **first**:
>
> ```
> #show: exo-page-columns.with(count: 2, rule: 0.5pt + gray)
> #show: exo-auto-chapter
> ```
>
> The other way round, the corrections that `exo-auto-chapter` prints at the end of the document sit outside the columned body and come out full width, silently. If you would rather not depend on the order, set `corr-columns` instead (see below): it wraps the corrections where they are printed, so it holds either way.

The rule is page furniture here: it is drawn across the full text area of every page, including a last page the exercises only half fill. That is deliberate for a document set in columns from end to end – the rule belongs to the page, not to the content – but it is the one place where `exo-columns`, whose grid stops the rule at the content, reads better. `rule-inset` trims a fixed amount off both ends:

```
#show: exo-page-columns.with(
  count: 3,
  gutter: 1cm,
  rule: 0.4pt + luma(160),
  rule-inset: 6pt,
)
```

14.2. One Columned Block

`exo-columns` puts a single block of the document on several columns while the rest stays full width – a page of drill exercises inside a chapter, for instance.

Code

```
#exo-columns(count: 2, rule: 0.5pt + gray, gutter: 0.7cm)[
  #exo(exercise: [Solve $2x + 3 = 13$.], solution: [$x = 5$])
  #exo(exercise: [Solve $3x - 6 = 0$.], solution: [$x = 2$])
  #exo(exercise: [Expand $(x + 1)^2$.], solution: [$x^2 + 2x + 1$])
  #exo(exercise: [Factor $x^2 - 9$.], solution: [$(x-3)(x+3)$])
]
```

Preview

Exercise 1

Solve $2x + 3 = 13$.

Solution 1

$x = 5$

Exercise 2

Solve $3x - 6 = 0$.

Solution 2

$x = 2$

Exercise 3

Expand $(x + 1)^2$.

Solution 3

$x^2 + 2x + 1$

Exercise 4

Factor $x^2 - 9$.

Solution 4

$(x - 3)(x + 3)$

Unlike page columns, a block has to know its height before `columns` can balance it – otherwise it fills the first column to the bottom of the region and leaves the others empty. By default `exo-columns` estimates that height by measuring the content at column width and dividing by `count`, plus `slack` lines of margin. The estimate is deliberately generous, since a block slightly too tall only ends with a shorter last column. It is also a cap: a body that outgrows it is cut off, and a body longer than a page is lost outright. Pass an explicit `height` to set the block's height yourself.

14.2.1. `balance: true`

`balance: true` takes a different route. Instead of `columns`, the body is split into its own pieces (one exercise is one piece) and handed to a grid, which draws the rule itself at the real height of the content and breaks across pages like any other block. Nothing is estimated and nothing is lost: the rule stops where the exercises stop, and a block longer than a page continues on the next one.

Code

```
#exo-columns(
  count: 2,
  rule: 0.5pt + gray,
  balance: true,   // exact; breaks across pages
)[
  #for i in range(1, 60) { exo(exercise: [...]) }
]
```

Preview

Exercise 1

Solve $2x + 3 = 13$.

Exercise 2

Solve $3x - 6 = 0$.

Exercise 3

Expand $(x + 1)^2$.

Exercise 4

Factor $x^2 - 9$.

It needs a rule to draw (a plain `columns` already balances and flows on its own, so there is nothing to gain), no explicit `height`, and a body it can take apart – a sequence of exercises, which is what `[ #for .. { exo(..) } ]` and a run of `#exo(..)` calls both are. A body that is one indivisible element – a `#block[..]` around everything, or a `set` rule applied to the whole body, whose styles could not be re-applied to the pieces one by one – silently falls back to the estimate, as does a body with fewer pieces than columns. Counter updates travel with the exercise they belong to, so a split never renumbers anything.

The columns are filled by piece count, not by measured height: every piece here is a `context` element, and `measure` cannot resolve an introspection against the real document, so measuring them would leave the document short of converging. One piece much taller than its neighbours therefore leaves its column longer than the others.

It is opt-in for one reason: two ruled grids in the same document – a split block **and** corrections in ruled columns (`corr-columns` with `corr-columns-rule`) – together with `exo-auto-chapter` can leave the document short of converging. One split block alongside ruled corrections is fine; several are not. If Typst reports "document did not converge", drop back to the default on the blocks that can afford the estimate.

For a document that is two-column from end to end, `exo-page-columns` is still the right tool.

14.3. Corrections on Two Columns

Collected corrections (`corr-loc: "end-chapter"` or `"end-section"`) are printed by `exo-print-solutions`, which is called from inside the flow rather than by you – so they cannot be wrapped in `exo-columns` by hand. `corr-columns` sets them on their own columns instead, whatever the rest of the document does:

```
#exo-setup(
  corr-loc: "end-chapter",
  corr-columns: 2,
  corr-columns-gutter: 0.7cm,
  corr-columns-rule: 0.5pt + gray,   // optional vertical bar
)
#show: exo-auto-chapter
```

Two layouts hide behind that setting, and they differ in how a long correction section behaves:

- **Without a rule**, this is Typst's own `columns`: the corrections flow from one column into the next and on across pages, exactly like body text.
- **With a rule**, they are laid out as a grid, which draws the vertical bar itself at the true height of the content on every page the section spans. The corrections are then distributed between the columns up front instead of flowing from one into the next, and by piece count rather than by measured height: every correction is a `context` element, and `measure` cannot resolve an introspection against the real document, so measuring them would leave it short of converging. A correction much taller than its neighbours therefore leaves its column longer than the others.

`corr-columns-badge-position` decides where the badge sits inside those columns: `auto` (the default) uses `"above"` as soon as there are two columns or more, since a badge column would eat a large share of a narrow measure. Pass `"margin"` to keep the badges beside the corrections anyway. `corr-columns-rule: none` removes a rule set earlier (that parameter's "leave as is" value is `auto`, unlike the rest of `exo-setup`, so that `none` can mean what it says).

For a document that is **entirely** two-column, use `exo-page-columns` instead and leave `corr-columns` at 1: the corrections then flow with everything else and the page rule is drawn for them too.

14.4. Badges in Narrow Columns

Both functions take `badge-position`, which they apply only to their own body and reset afterwards:

- `auto` (the default) uses `"above"`: the badge sits on a header line and the statement runs the full column width. This is what makes two columns readable – see Section 13.13.
- `none` leaves the current configuration untouched, badges keep whatever `exo-setup` set.

- Any explicit value (`"margin"`, `"above"`) is used as given.

The badge styles that wrap the whole exercise (`border-accent`, `underline`, `rounded-box`, `header-card`) already run the full width and are unaffected. The exception is `margin`, whose side label is a fixed column of its own: in a two-column measure it would leave the statement about as wide as the label. It therefore folds – the rule runs the whole measure, the label sits under it on the right, the statement keeps the full width below – either when the measure drops under `margin-fold-below` (`auto` = three times the label column) or whenever `badge-position` is `"above"`, which is exactly what the two column functions ask for. `margin-label-width` and `margin-label-gutter` set the unfolded geometry; `margin-fold-below: 0pt` never folds.

15. Parameter Reference

15.1. `exo` Function

Parameter	Type	Default	Description
`exercise`	content	none	Exercise content (required)
`solution`	content	none	Solution content
`correction`	content	none	Correction for teachers
`id`	string	auto	Unique exercise ID
`margin-content`	content	none	Content placed below the badge (e.g. remarks)
`qr`	string/content	none	QR code for the exercise box (URL string or content)
`qr-sol`	string/content	none	QR code for the solution box
`qr-corr`	string/content	none	QR code for the correction box
`sol-in-corr`	bool	false	If true, solution is already in correction (don't show both)
`show-corr`	bool	false	If true, show correction in "mixed" mode
`optional`	bool	false	Show the optional marker before the label
`corr-given`	bool	false	Show the correction-given marker (dumbbell icon)
`topic`	string	none	Topic metadata
`level`	string	none	Difficulty level
`authors`	array	()	Author names

15.2. `exo-define` Function

Same as `exo`, plus:

Parameter	Type	Default	Description
`optional`	bool	false	Show the optional marker before the label
`corr-given`	bool	false	Show the correction-given marker (dumbbell icon)
`competencies`	array	()	Competency tags
`points`	number	none	Points (for exam mode)

15.3. `exo-select` Function

Parameter	Type	Default	Description
`topic`	string	none	Filter by topic
`level`	string	none	Filter by level
`author`	string	none	Filter by author
`competency`	string	none	Filter by competency
`topics`	array	none	Filter by any topic

`levels`	array	none	Filter by any level
`competencies`	array	none	Filter by any competency
`where`	function	none	Custom filter function
`renumber`	bool	true	Renumber sequentially
`max`	int	none	Maximum exercises

15.4. `exo-setup` Function

Display Control:

Parameter	Type	Default	Description
`display`	string	"both"	"ex", "sol", "both"
`corr-display`	string	"solution"	"solution", "correction", "mixed"
`corr-loc`	string	"after"	"after", "pagebreak", "end-section", "end-chapter"
`sol-loc`	string/auto	auto	Same values as `corr-loc`, for solutions only
`exercise-label`	string	"Exercise"	Label for exercises
`solution-label`	string	"Solution"	Label for solutions
`correction-label`	string	"Correction"	Label for corrections
`counter-reset`	string	"section"	"section", "chapter", "global"
`number-prefix`	none/str/counter/function	none	"heading" (level-1 heading number), a custom counter, or a function
`number-separator`	string	"."	Separator for chapter-prefixed numbers
`show-id`	bool	false	Show exercise IDs
`show-competencies`	bool	false	Show competency tags
`draft-mode`	bool	false	Show placeholders for empty content
`solution-placeholder`	content	*To be completed*	Placeholder text
`correction-placeholder`	content	*To be completed*	Placeholder text

Visual Styling:

Parameter	Type	Default	Description
`badge-style`	string/function	"box"	Style: "box", "circled", "filled-circle", "rect", "filled-rect", "pill", "tag", "margin", "border-accent", "underline", "rounded-box", "header-card" – or a custom badge function
`badge-position`	string	"margin"	"margin" (badge in its own left column) or "above" (badge on a header line, statement full width below)
`badge-color`	color	black	Color for exercise badges

`solution-color`	color	green	Color for solution badges
`correction-color`	color	green	Color for correction badges
`label-font-size`	length	12pt	Font size for badge labels
`badge-scale`	float	1.0	Multiplies the built-in badge paddings and the circle diameter (the label text keeps `label-font-size`)
`badge-pad-x`	length/auto	auto	Horizontal padding inside the badge, overriding `badge-scale`
`badge-pad-y`	length/auto	auto	Vertical padding; the badge is `label-font-size + 2 * badge-pad-y` tall
`badge-radius`	length/auto	auto	Corner radius of the badge (auto = the style's own, scaled)
`margin-label-width`	length	3.35cm	Side-label column of badge-style "margin"
`margin-label-gutter`	length	0.55cm	Gap between that column and the statement
`margin-fold-below`	length/auto	auto	Below this measure the "margin" side label folds onto a header line (auto = 3 × `margin-label-width`; 0pt = never)
`margin-position`	length/auto	auto	Width reserved for badge column
`label-extra`	length	1cm	Extra space for labels in margin

Spacing:

Parameter	Type	Default	Description
`exercise-above`	length	0.8em	Space above exercise boxes
`exercise-below`	length	0.8em	Space below exercise boxes
`solution-above`	length	0.8em	Space above solution boxes
`solution-below`	length	0.8em	Space below solution boxes
`correction-above`	length	0.8em	Space above correction boxes
`correction-below`	length	0.8em	Space below correction boxes
`advanced-symbol`	content/none	`"*"`	Symbol before label for advanced exercises
`optional-symbol`	content/none	star icon	Symbol before label for optional exercises
`corr-given-symbol`	content/none	dumbbell icon	Symbol before label when correction is handed out

Difficulty, Inline Solutions, and Links:

Parameter	Type	Default	Description
`difficulty-display`	string	"color"	"color", "stars", "symbols", "none"

`difficulty-scale`	auto/dict	auto	auto = built-in 5-level scale, or dict key -> (color: .., symbol: ..)
`difficulty-position`	string	"below"	Stars/symbols "below" the badge or inline in the "badge"
`solution-style`	auto/string	auto	"inline" = epigraph-style rule instead of a badge box
`inline-rule-length`	length	3cm	Rule length for inline solutions
`inline-label`	auto/content/none	auto	Small margin label for inline solutions
`corr-columns`	int	1	Number of columns for the collected (end-section / end-chapter) corrections
`corr-columns-gutter`	length/ratio	4%	Gutter between those columns
`corr-columns-rule`	stroke/none	none	Vertical separator between them, e.g. `0.5pt + gray`
`corr-columns-badge-position`	auto/string	auto	Badge position inside those columns; auto = "above" from two columns on
`link-solutions`	bool	false	Clickable links to deferred corrections
`link-style`	string	"icon"	"icon" (arrow) or "page" ("Solution p. 30" reference)
`link-icon`	content/none	arrow icon	Link icon on the exercise
`backlink-icon`	content/none	arrow icon	Back-link icon on the correction
`page-ref-format`	auto/function	auto	Custom page reference: (label, page) => content
`page-ref-color`	auto/color	auto	Page reference color (auto = badge color)

QR Code Settings:

Parameter	Type	Default	Description
`show-qr`	bool	true	Master toggle for per-exercise QR codes
`qr-size`	length	1.5cm	Target QR code size (shrinks to fit margin if needed)
`qr-min-size`	length	1cm	Minimum size; below this the QR extends into page margin
`qr-color`	color	black	QR module color
`qr-caption`	content/none	none	Small caption below every QR code
`qr-position`	string	"auto"	"auto" (per badge style), "wrap" (always wrap content), or "tasks" (overlay; a taskize `#tasks` body flows around it)

15.5. `exo-page-columns` Function

Show rule: `#show: exo-page-columns.with(...)`.

Parameter	Type	Default	Description
`count`	int	2	Number of page columns
`gutter`	length | ratio	4%	Space between columns
`rule`	stroke	none	Vertical separator drawn in each gutter
`rule-inset`	length	0pt	Shortens the rule at top and bottom
`badge-position`	str | auto | none	auto	auto = "above" inside the columns; none = leave the configuration alone

15.6. `exo-columns` Function

Same parameters as `exo-page-columns`, plus the height controls of the block:

Parameter	Type	Default	Description
`balance`	auto | bool	auto	auto (default) = flow through `columns` in a block of estimated height; true = split the body between the columns, which is exact and breaks across pages (needs a `rule`, no explicit `height`, and a splittable body)
`height`	length | auto	auto	Fixed block height; implies the flowing path
`slack`	int	3	Lines of margin added to the estimated column height (flowing path only)

`rule-inset` applies to the flowing path only: the split path's grid already stops the rule at the content.

16. Utility Functions

Function	Description
`exo-reset-counter()`	Reset exercise numbering to 0
`exo-clear-registry()`	Clear all registered exercises
`exo-get-registry()`	Retrieve the current registry of exercise metadata (context-dependent)
`exo-section-start()`	Call at section start (triggers reset if configured)
`exo-chapter-start()`	Call at chapter start (triggers reset if configured)
`exo-section-end()`	Print solutions pending for end-of-section placement
`exo-chapter-end()`	Print solutions pending for end-of-chapter placement
`exo-print-solutions()`	Print collected solutions (for end-section/chapter modes)
`exo-count(topic: ..)`	Count exercises matching criteria
`exo-show("id")`	Display exercise by ID
`exo-show-many("a", "b")`	Display multiple exercises by ID
`exo-filter(topic: ..)`	Filter and display matching exercises
`competency-tag("comp")`	Render a single competency as a small styled pill (used internally by `show-competencies`)

17. Exam Integration

These functions support building exam documents from an exercise bank, typically used alongside a layout package such as `texam`.

17.1. `exam-setup`

Configure the exam correction box appearance and behaviour.

Parameter	Type	Default	Description
`show-solutions`	bool	`false`	Show `exam-solution-box` content
`solution-label`	string	`"Correction"`	Label shown in bold at the top of the solution box
`solution-fill`	color	green tint	Background fill of the solution box
`solution-stroke`	stroke	green stroke	Border stroke of the solution box
`solution-radius`	length	`3pt`	Corner radius of the solution box
`solution-inset`	length	`8pt`	Inner padding of the solution box
`solution-label-color`	color	dark green	Color of the label text
`question-spacing`	length	`1em`	Vertical space between questions

17.2. `exam-solution-box`

A green-highlighted answer block. Only rendered when `exam-setup(show-solutions: true)`.

```
#exam-solution-box[
  $(x - 2)(x - 3)$
]
```

Place it after an exercise. The same source file produces both the student version (box hidden) and the correction (box shown) by flipping `show-solutions`.

17.3. `exam-score-table`

Builds a score grid from registered exercise IDs, reading point values from the registry.

```
#exam-score-table("ex-1", "ex-2", "ex-3")
```

Parameter	Type	Default	Description
`..exercise-ids`	strings		Positional — IDs of exercises to include
`question-label`	string	`"Question"`	Header for the question column
`points-label`	string	`"Points"`	Header for the points row
`grade-label`	string	`"Note"`	Header for the grade row
`total-label`	string	`"Total"`	Label for the total column